

1	(a)	The work done per unit mass , by an external force, in bringing a small test mass from infinity to that point , without any change in kinetic energy.		B1
	(b)	(i)	As the satellite goes to a higher orbit, gravitational potential energy , $U = -\frac{GMm}{R}$ will become less negative as R increases, hence the gravitational potential energy of the satellite increases .	B1
		(ii)	The gravitational force between the Earth and the satellite provides for the centripetal force that keeps the satellite of mass m in its circular orbit. $\frac{GMm}{R^2} = \frac{mv^2}{R}$ $v = \sqrt{\frac{GM}{R}}$	B1 A0
		(iii)	Since G and M are constant, it follows from part (ii) that $v \propto \sqrt{\frac{1}{R}}$ Hence, $v_{\text{new}} = \sqrt{\frac{R_{\text{old}}}{R_{\text{new}}}} v_{\text{old}} = \sqrt{\left(\frac{6610 \times 10^3 \text{ m}}{6890 \times 10^3 \text{ m}}\right)} (7780) \text{ m s}^{-1}$ $= 7620 \text{ m s}^{-1}$	M1 A0
		(iv)	1. (Change in kinetic energy) = (final kinetic energy) – (initial kinetic energy) $= \frac{1}{2} (120) 7620^2 - \frac{1}{2} (120) 7780^2$ $= -1.48 \times 10^8 \text{ J}$	M1 A1
		2.	Let KE be the kinetic energy, GPE the gravitational potential energy, and TE the total energy. Using part (ii), we have $\text{KE} = \frac{1}{2} mv^2 = \frac{1}{2} \frac{GMm}{R}$ Furthermore, we have $\text{GPE} = -\frac{GMm}{R}$ Hence, $\text{TE} = \text{KE} + \text{GPE} = \frac{1}{2} \frac{GMm}{R} + \left(-\frac{GMm}{R}\right) = -\frac{1}{2} \frac{GMm}{R}$ Since TE = -KE (needs to be derived) , (change in total energy) = (negative of change in kinetic energy) $= +1.48 \times 10^8 \text{ J}$	M1 A1
			Total:	8

2023

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2	(a)	(i)	N : the number of molecules in the ideal gas	
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